

PANIC SELLING AND INVESTOR BEHAVIOR IN THE INDIAN STOCK MARKET

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**MASTER OF SCIENCE IN ECONOMETRICS AND FINANCIAL
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DECLARATION

I, NIKHITHA VINEETH, do hereby declare that the project titled “Panic Selling and Investor Behavior in the Indian Stock Market” has been undertaken by me as part of the second semester of the Master of Science in Econometrics and Financial Technology, under the mentorship of Mr. Sachin P., Senior Manager, Rubrix Analytics Pvt. Ltd.

I also declare that this project contains no more than an academically acceptable level of similarity and has not been submitted before for the award of any degree, diploma, associateship or fellowship or any other title in University or elsewhere.

Place: Kochi

Nikhitha Vineeth

Date: 30.04.2025

Reg. No: 24030098

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Place: Cochin

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NIKHITHA VINEETH

ABSTRACT

Fear and uncertainty is observed in the stock market during market crashes. This fear and uncertainty leads to panic selling of securities which is seen as a concern. This project examines how the panic selling behavior causes disruptions in the Indian stock market, focusing on the timeframe between 2014 and 2024. Using the daily data collected from the National Stock Exchange, multiple regression model was applied for the analysis to check how the trading volume depends on the volatility and prices of the stock. From the analysis, it is seen that the trading volume is significantly impacted by both the volatility and prices of the stock. High kurtosis and skewness were seen in results of trading, suggesting the presence of extreme irrational behavior among the investors. Using the charts, it enabled to visually see the trends in the volatility during the depicted timeframe, as well as seeing the increasing trend of the stock prices.

Keywords: Panic Selling, Stock Market, Volatility, Trade Volume

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I. INTRODUCTION

The stock markets are steered by two powerful emotions of investors which are fear and greed. These emotions influence the investors to make irrational decisions. One such irrational behavior is the panic selling. Panic selling is a situation where large scale selling of securities takes place in an uncoordinated manner due to sharp decline in prices. It is mainly triggered due to major negative news, economic shocks and market crashes.

In India, several events had happened between 2014 and 2024 that had caused panic selling episodes in the Indian stock market. Major macroeconomic events like the Indian Rupee Demonetisation has caused Nifty 50 fell by 541 points intraday. Other countries policies and events also affects the Indian stock market like the China Yuan devaluation which has caused Nifty 50 to fall by 491 points and Sensex by 1624 points. Global events like the COVID-19 pandemic had a huge impact on all the stock markets across the globe, through which India faces significant market crashes causing the volatility index surged to over 83, which is all time high, indicating extreme fear.

Company or industry specific events can also have impact on the stock markets, especially the conglomerate firms. An evident example in present is the allegations against the Adani Group that started happening during January 2023 which caused the Sensex and Nifty 50 to fell sharply over multiple sessions due to heavy selling in the Adani stocks. Thus, Panic selling is an interesting phenomenon that needs to be analysed to know its impact on the stock market as well as on the Indian economy.

II. LITERATURE REVIEW

Gupta (2019) analysed herding behavior that arise due to panic selling using cross-sectional standard deviation and absolute deviation of stock returns. The study finds this extreme herding behavior on bull phase of market and highlights on investor biases causing changes in the decision making process.

Barlevy and Veronesi (2003) emphasized that stock market crashes can impact the stock even when the company has strong fundamentals. The study also says that the market crashes are primarily done by retail investors as they don't have access to inside or informed information.

Aliyu Ali Bawalle (2025) analyses the investor's overreaction to bad news and how it leads to panic selling. The study also incorporates financial literacy as a variable to their analysis. The results shows that overconfident investors were more likely to panic sell their stocks on bearish phases of market, and on how financial literacy helped to reduce panic selling to a certain extent.

III. RESEARCH GAP

Most of the literature available presently is purely descriptive in nature. There aren't any analytical papers on how to evaluate the panic selling. Most of the articles explain about the behavioral biases rather than delving into the technical parts on how to analyse the panic selling episodes.

IV. OBJECTIVE

To analyse the occurrence of panic selling episodes in Indian stock market happened between 2014-2024

V. METHODOLOGY

Data :

The data for conducting the analysis is based on secondary data which is available in the official website of NSE. The dataset includes the daily data of closing price of Nifty 50, close index of VIX, and shares traded from 1st January 2014 to 31st December 2024.

Model:

To analyse the panic selling, the variables under study are the trading volume of the shares, close index of the volatility index and the natural log of the closing prices of the stock. To fit it in a regression model, we analyse how trading volume is affected by the volatility and stock prices:

$$ST_t = -\beta_0 + \beta_1 P_t + \beta_2 V_t + \epsilon_t$$

Where,

- ST_t is shares traded
- P_t is price log of the stock
- V_t is volatility index

VI. ANALYSIS AND INTERPRETATION

The dataset has been cleaned and arranged in a systematic manner for conducting the analysis. The descriptive statistics is implemented on the dataset to know about the characteristics of the variables under study. Then multiple regression analysis was done to know how the trading volume depends on the volatility index and price of the stock.

Table 1 : Descriptive Statistics

	<i>Shares traded</i>	<i>VIX</i>	<i>Price Log</i>
Mean	3.13E+08	16.99579	9.397232
Median	2.54E+08	15.62000	9.312527
Maximum	1.81E+09	83.60750	10.17413
Minimum	9774392	10.13500	8.699665

Std. Dev.	1.89E+08	6.216271	0.373451
Skewness	1.975069	4.156262	0.319731
Kurtosis	9.006989	31.20521	1.940048
Jarque-Bera	5851.456	98027.61	173.7369
Probability	0.000000	0.000000	0.000000
Sum	8.50E+11	46245.54	25569.87
Sum Sq. Dev.	9.73E+19	105106.3	379.3458
Observations	2717	2717	2717

Source: Author's Calculation

Looking onto the descriptive statistics of the dataset, it can be seen that the variables shares traded and VIX are leptokurtic, as they are skewed towards right whereas the price shows platykurtic. The minimum and maximum values of VIX shows a wide dispersion. The standard deviation for shares traded is very much high compared to VIX and price log, showing the high variability. None of the variables come closer to normality as Jarque-Bera test shows extreme non normal values.

Table 2 : OLS

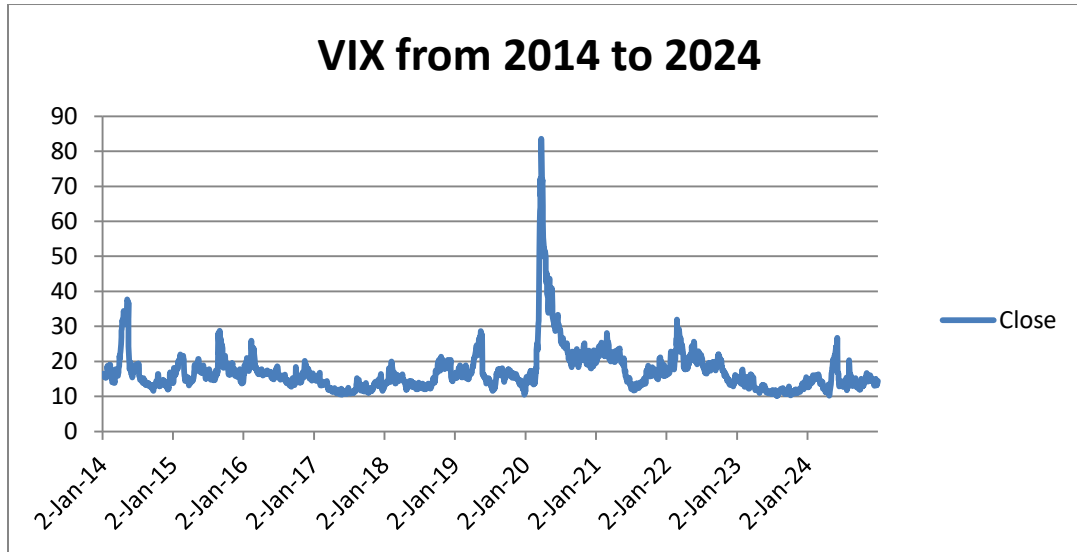
<i>Variable</i>	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-statistic</i>	<i>Prob.</i>
β_0	-1.28E+09	80276538	-15.99862	0.0000 ***
P_t	1.42E+08	8329860	17.00573	0.0000 ***
V_t	15641579	499695.2	31.30224	0.0000 ***
R-squared		0.291314	Mean dependent var	3.13E+08
Adjusted R-squared		0.290791	S.D dependent var	1.89E+08
S.E of regression		1.59E+08	Akaike Info Criterion	40.61313
Sum of squared Residuals		6.90E+19	Schwarz criterion	40.61549
Log likelihood		-55169.94	Hannan-Quinn Criter.	40.61549
F-statistic		557.8103	Durbin-Watson stat	0.428232
Prob(F-statistic)		0.000000		

Source: Author's Calculation

*** indicates $P < 0.05\%$

Based on the results, the model is overall significant as the predictors are statistically significant (p values are less than 0.001.). the intercept, which is β_0 , is statistically significant as when volatility and price are zero, shares traded would be -1.28, which theoretically doesn't make sense. Looking onto R^2 , the model explains about 29% variations in shares traded which is low. The Durbin-Watson statistic shows 0.42 which shows the presence of positive autocorrelation.

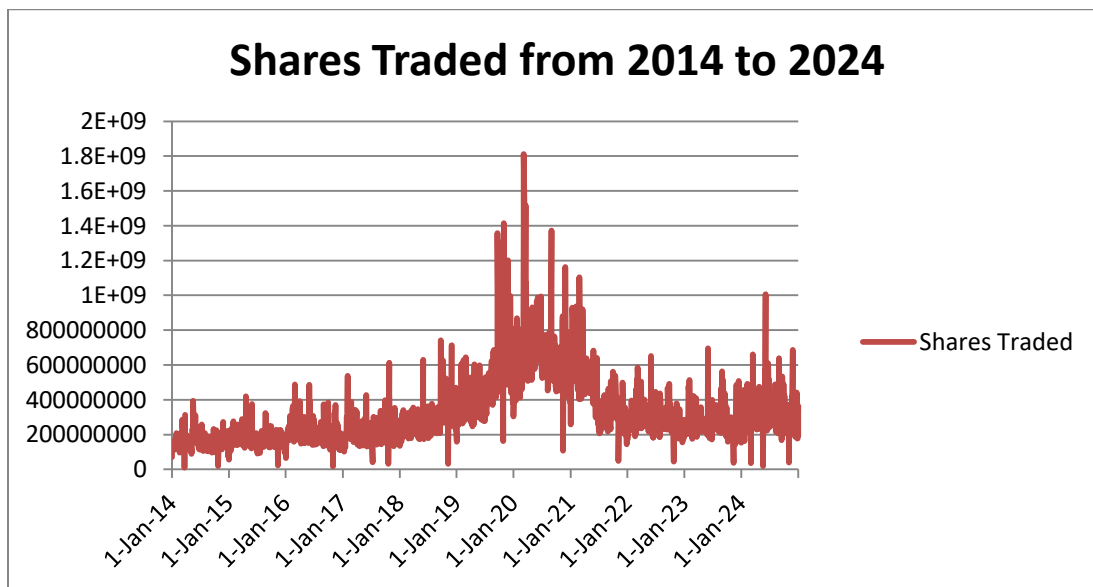
Figure 1 : VIX from 2014- 2024



Source: Author's Calculation

From this line chart, it can be evidently seen the vix surged to its highest value during 2020, which is 83 due to COVID-19 pandemic. Here, the panic selling was at peak as there was economic downturn leading the investors to sell their stocks to avoid making losses.

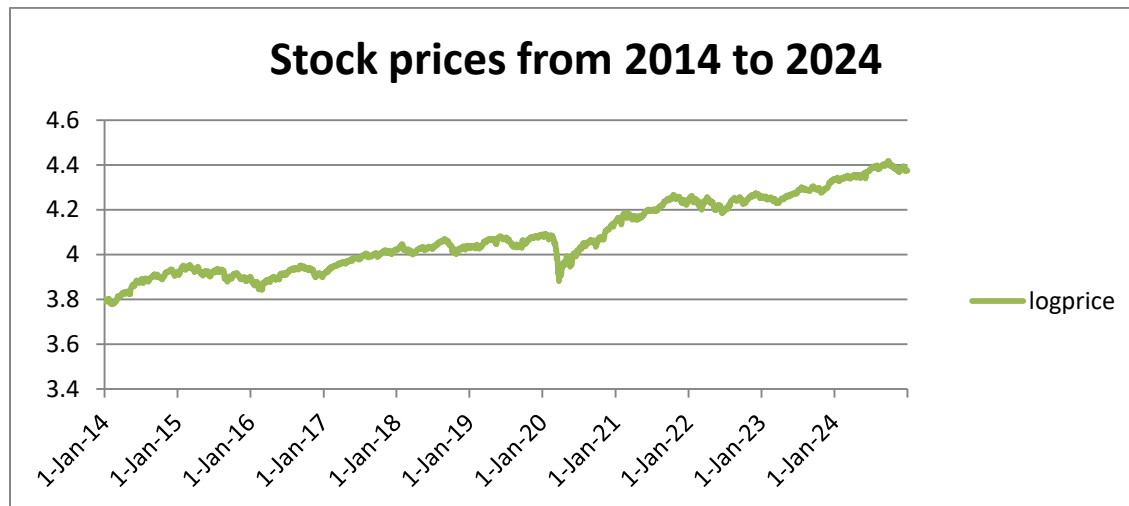
Figure 2: Shares Traded from 2014- 2024



Source: Author's Calculation

In chart 2, Considering the pandemic happened the volume of shares traded peaked during 2020, making the investors to sell their securities causing herd behavior where the investors follows the crowd in trading their stocks.

Figure 3: Stock prices from 2014- 2024



Source: Author's Calculation

From chart 3, we can see the increasing trend in the share prices during the decade, indicating that the panic selling episode is temporary, lasting mostly over a year at the maximum.

VII. CONCLUSION

This study overall identifies and examines the panic selling behavior during the chosen decade. Using the available data, the analysis has found that both volatility and stock prices have a positive impact on the volume of shares traded, especially prices shows a significant impact with high t-statistics and low p value. Moreover high volatility is explained through extreme values which was shown in the analysis stating high emotional trading had happened. High skewness, kurtosis and non normality in the dataset has indirectly shown the irrational behaviors of the investors during the given timeframe. Thus, we can say that investors are not rational and panic selling has a huge impact on the stock market as well as for the wellbeing of the economy.

VIII. REFERENCES

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